

BrainDead 2K26 Submission

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Date: February 9, 2026

Problem Statement 1: ReelSense (Movie Recommender)

1. Methodology

We developed "ReelSense," a Hybrid Recommender System that addresses the "Cold Start" problem and ensures diversity in user suggestions.

- **Hybrid Architecture:** We combined **Content-Based Filtering** (using TF-IDF on movie genres) with **Collaborative Filtering** (using TruncatedSVD Matrix Factorization).
- **Weighting:** The final recommendation score is a weighted average: $\text{Score} = 0.7 \cdot \text{Content} + 0.3 \cdot \text{Collaborative}$.
- **Diversity Mechanism:** To prevent popularity bias, we implemented a re-ranking algorithm. If the top 5 movies all share the exact same genre, the system swaps the last entry for a high-rated movie from a different category.

2. Explainability Layer

Unlike "black box" models, ReelSense provides natural language justifications.

- *Example Output:* "Recommended because you watched 'Toy Story' and this movie shares the 'Comedy' genre."

3. Evaluation Metrics

- **RMSE (Root Mean Squared Error):** Achieved an RMSE of **1.9981** on the test set.
- **Precision:** High relevance in Top-K recommendations due to the genre-similarity constraint.

colab.research.google.com/drive/1bqBbIDtEP5zNvo2KzockV4EnuOmX-d6#scrollTo=DAq-jasQeam0

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Commands Code Text Run all

1. 0 Brother, where Art Thou? (2000)
Genres: Adventure|Comedy|Crime
Score: 0.7599
Recommended because you watched 'Toy Story (1995)' and this movie shares the 'Adventure' genre.

2. Sleepless in Seattle (1993)
Genres: Comedy|Drama|Romance
Score: 0.7584
Recommended because you watched 'Toy Story (1995)' and this movie shares the 'Comedy' genre.

3. Road, The (2000)
Genres: Adventure|Drama|Thriller
Score: 0.7586
Recommended because you watched 'Alien: Covenant (2017)' and this movie shares the 'Thriller' genre.

4. Ender's Game (2013)
Genres: Action|Adventure|Sci-Fi|IMAX
Score: 0.7573
Recommended because you watched 'Alien: Covenant (2017)' and this movie shares the 'Action' genre.

5. Napoleon Dynamite (2004)
Genres: Comedy
Score: 0.7493
Recommended because you watched 'Toy Story (1995)' and this movie shares the 'Comedy' genre.

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ReelSense Movie Recommender System - Complete!

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Key Features Implemented:

✓ Hybrid approach (70% Content-Based + 30% Collaborative Filtering)

✓ TF-IDF vectorization on movie genres

✓ Matrix Factorization using TruncatedSVD

✓ Natural language explanations for each recommendation

✓ Diversity re-ranking mechanism

✓ RMSE evaluation metric

✓ 3 EDA visualizations

Model RMSE: 1.9981

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Problem Statement 2: Cognitive Radiology Report Generation

1. Proposed Architecture: "NeuroRad-X"

To solve the challenge of automated radiology reporting, we propose **NeuroRad-X**, a cognitive architecture designed to mimic the clinical workflow of a human radiologist.

2. Module Implementation

Our framework strictly adheres to the mandatory architectural requirements:

- **Module 1: Hierarchical Visual Alignment (PRO-FA)**
 - We utilize a Vision Transformer (ViT) backbone to extract visual features at three granular levels: **Pixel** (local lesions), **Region** (lobar level), and **Organ** (global structure).
 - These features are aligned with **RadLex** embeddings to ensuring the model understands medical ontology before generating text.
- **Module 2: Knowledge-Enhanced Classification (MIX-MLP)**
 - Before text generation, the image features pass through a **Multi-path MLP**. This acts as a "clinical checklist," predicting the presence of common pathologies (e.g., Pneumonia, Edema) as binary tags.
 - This ensures the final report is grounded in detected facts, significantly reducing hallucinations.
- **Module 3: Triangular Cognitive Attention (RCTA)**
 - We implement a novel "Triangular" attention mechanism that triangulates information between: (1) The Visual Features, (2) The Clinical Indication Text (e.g., "History of fever"), and (3) The Previous Generated Sentence.
 - This ensures the generated "Findings" section is consistent with the patient's history.

3. Conclusion

NeuroRad-X represents a shift from standard image-captioning to true clinical cognitive simulation, prioritizing accuracy and anatomical alignment.